

RECOVERING HIDDEN ORDER IN SATURATED DIFFICULTY RATINGS

Jhen-Ke Lin

National Yang Ming Chiao Tung University
jacob.cs14@nycu.edu.tw

ABSTRACT

Rhythm-game players choose challenges by difficulty rating, but as harder charts join a fixed scale, many share its highest rating. A model could provide a finer order, but must learn from tied labels that cannot confirm it. We propose artificial top-coding, adapted from artificial censoring, which merges known upper ratings during training to recreate the tie where the order is known. On *Taiko no Tatsujin*, gradient-boosted trees lead seven chart-level models in full-scale correlation yet rank the merged 8–10 charts worst. Treating the merged rating as a lower bound rather than an exact target, with only the loss changed, improves ranking for chart-level and sequence models. For charts actually tied at the top, lower-bound models’ scores also agree with independently designed expert-course levels. Ceiling labels should therefore be learned as lower bounds, and rankings within them tested by withholding known levels.

Index Terms— difficulty estimation, ordinal regression, censored regression, ranking evaluation, symbolic music

1. INTRODUCTION

A rhythm-game chart specifies the timed actions a player must perform. Difficulty ratings help players choose charts, but a fixed rating scale can lose resolution as harder content is added. In *Taiko no Tatsujin*, the expert-oriented Oni chart category has an official scale from 1 to 10 [1]. Many expert charts receive 10 even though they appear at different stages of the game’s advanced courses. This is a ceiling effect: the highest available rating groups items that may differ in difficulty [2].

A model could provide a finer order for these charts, but two difficulties arise. Its training labels are tied at the ceiling, so they give the model no order to learn there, and the shared rating cannot establish whether a predicted order is correct. Agreement across the full scale may mostly reflect how well a model separates the lower levels. This resembles the range effect in image-quality assessment, where strong correlation across a broad range can coexist with weak discrimination within a narrow range [3]. At a rating ceiling, the problem is stronger: all reference ratings in the group are identical.

Difficulty models for rhythm games and piano use chart statistics, note sequences, and ordered classes [4–7]. Franks

et al. [6] and Ramoneda et al. [7] compare predicted rankings with published labels, player judgments, or publisher orderings. Ordinal regression [8–10] and interval-label learning [11, 12] model coarse labels. Closest to our test, Dănilă and Buiu [13] artificially censor outcomes for training and validation, then score each prediction against its original value, as PRIL [12] does after training on intervals. These studies score whole-scale rankings or each censored value, not order within a shared top rating.

Our key observation is that this tie can be recreated where the answer is known: merging known upper levels into one rating, an operation called *top coding*, hides their order during training and model selection, while the original ratings of held-out charts still record it. The merged rating is then a lower bound, as in Tobit models [14]. Motivated by this, we propose a method that makes order at the ceiling verifiable, and use it to find how that order should be learned. Our contributions are: (i) *Method*. Artificial top-coding adapts artificial censoring [13]: ratings 8, 9, and 10 all become 8 during training and model selection, and test scores for this *capped group* are checked against the original ratings (Sec. 2.1). (ii) *Principle and mechanism*. Merged ratings should be learned as lower bounds, not exact targets; an exact target pulls capped charts toward the cap, where their order is lost (Secs. 2.2 and 4.2). (iii) *Evidence*. The model with the highest full-scale correlation ranks the capped group worst; treating merged ratings as lower bounds improves ranking when only the loss changes, with chart-level features and a shared Transformer; and for the real ceiling, scores for 10-rated charts agree with expert-course levels, an external order (Sec. 4).

2. METHOD

2.1. Artificial top-coding

For an original rating $y_i \in \{1, \dots, K\}$, we define the observed rating z_i and the capped group H_c at the chosen cap c as

$$z_i = C_c(y_i) = \min(y_i, c), \quad H_c = \{i : y_i \geq c\}. \quad (1)$$

Every chart in H_c has the same observed rating c . Permuting their original ratings therefore leaves the observed rating vector unchanged. Any evaluation using only that vector is invariant to the hidden permutation: it cannot verify order

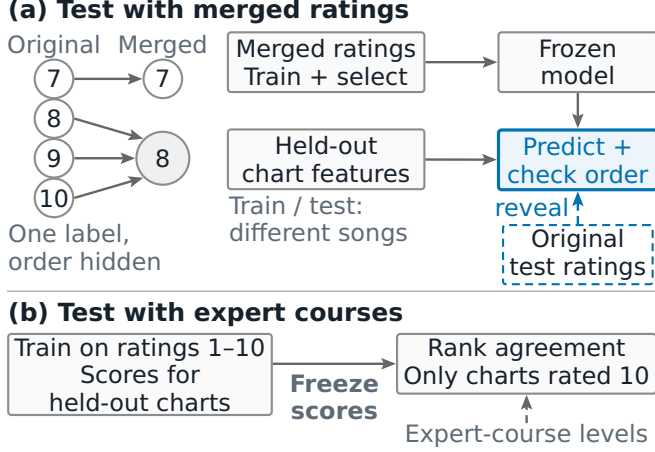


Fig. 1. Two tests of ranking at a rating ceiling. (a) Merging known upper levels at a chosen cap hides their order during training and model selection; the original ratings of held-out charts then check the predicted order. (b) Separately, scores from models trained on the original 1–10 scale are compared with expert-course levels for 10-rated charts.

within H_c . Chart features may still predict differences, but checking those predictions requires information beyond the shared rating. Artificial top-coding therefore applies Eq. (1) at a cap below the real ceiling, where original ratings are known.

Figure 1 summarizes the tests. We split charts into five test folds by normalized song title, keeping all versions of a song together. For each fold, the remaining charts supply training data and a separate validation split, also grouped by title. Training, hyperparameter selection, and early stopping use only z_i . Validation uses mean absolute error (MAE) after clipping expected ratings to $[1, c]$, so predictions above the cap are equivalent for selection. Original ratings above the cap and course information become available only after model selection. Each chart’s *test score* is an out-of-fold prediction by a model trained without its song-title group.

We evaluate test scores s_i within H_c using Spearman’s ρ , Kendall’s τ_b , and *pair accuracy*

$$A_c = |\{(i, j) \in P_c : s_i < s_j\}| / |P_c|, \quad (2)$$

the fraction of pairs $P_c = \{(i, j) : i, j \in H_c, y_i < y_j\}$ with unequal original ratings that the scores order correctly. Prediction ties receive no credit. These measures evaluate order; score gaps are not interpreted as star units.

2.2. Learning ceiling labels as lower bounds

For an artificial cap, $z_i = c$ implies $y_i \geq c$. Ordinary Huber fits s_i to c ; one-sided Huber penalizes only $s_i < c$, as in censored regression [13, 14]. With Huber penalty h_δ ,

$$\ell_{1s}(s, z; c) = \begin{cases} h_\delta(s - z), & z < c, \\ \phi(c - s), & z = c. \end{cases} \quad (3)$$

The global solver uses $\phi(u) = \max(u, 0)$; the Transformer uses $\phi(u) = h_\delta(\max(u, 0))$. Both penalties are zero when $s \geq c$; on the original scale, $c = 10$. For $z < c$, Eq. (3) equals the ordinary Huber loss, so the two differ only on capped charts. Ordinal models are instances of the same principle. Cumulative logit and probit models place ordered thresholds on a latent score [8]; they and ordinal reduction into nested binary classification tasks impose no upper target on capped charts.

An exact target pulls predictions for capped charts toward c ; the loss treats any spread among capped charts as error, so order within the ceiling is flattened. A lower bound does not penalize predictions above c , and ordinal likelihoods keep rewarding confidence in the top level, so charts whose features indicate greater difficulty still score higher, and order is preserved.

3. EXPERIMENTAL SETUP

3.1. Data and representations

We use 1023 unique Oni charts from 1019 song-title groups. Each chart contains drum-action types and timings, tempo, scroll speed, and sustained actions. These symbolic event sequences provide both chart-level summary features, called *global features* below, and inputs to a Transformer. The 19 global features summarize duration, note density, time between onsets, tempo, scroll speed, and note types. The sequence model encodes drum-action type, onset within the measure, sustained-action duration, tempo, scroll speed, and nearby events per second. Two- and four-measure windows start every two measures, with at most 128 events per window and 64 windows per chart.

3.2. Models and training

We compare seven global models: ordinary Huber regression, one-sided Huber regression, ridge regression, gradient-boosted decision trees (GBDT), cumulative logit and probit models, and ordinal reduction. Ordinary Huber, ridge, and GBDT fit the cap as an exact target; the others treat it as a lower bound. All global models use the same training, validation, and test partitions.

A four-layer, 256-dimensional Transformer [15] encodes each window; averaging valid events and windows gives the chart representation. Ordinary Huber, one-sided Huber, and cumulative logit share the encoder, partitions, optimizer, and training budget; only the output layer and loss change. At cap 8, all three sequence models use five folds and seeds 2025–2027. Caps 7 and 9 use prespecified single-seed ordinary Huber and cumulative logit runs. The full-scale sequence model uses cumulative logit with three seeds. After selecting an epoch, we refit on the fold’s training charts. Scores are standardized using that training set before combining folds

Table 1. Spearman correlation with original ratings. Full scale: all charts, after training on ratings 1–10. Cap columns: charts originally rated at or above the cap, after training on merged ratings; cap 8 (699 charts) is the main comparison. Upper rows use global features; Transformer rows differ only in output layer and loss; each Huber pair, only in the loss. GBDT is highest on the full scale but lowest at caps 7 and 8. Brackets: SD over three seeds; other sequence results use one seed. Dashes: not run.

Model	Full scale	Cap 7	Cap 8	Cap 9
Huber (ordinary)	0.881	0.493	0.640	0.636
Huber (one-sided)	0.883	0.722	0.732	0.679
Ridge	0.876	0.471	0.589	0.587
GBDT	0.897	0.349	0.435	0.599
Cumulative logit	0.878	0.811	0.761	0.693
Cumulative probit	0.880	0.809	0.760	0.690
Ordinal reduction	0.890	0.795	0.766	0.742
Transformer + ordinary Huber	—	0.415	0.454 [0.033]	0.475
Transformer + one-sided Huber	—	—	0.671 [0.010]	—
Transformer + cumulative logit	0.885 [0.001]	0.698	0.718 [0.003]	0.701

and seeds. Hyperparameter grids, optimizer settings, per-seed results, and run status are in the accompanying artifact.

Cap 8, fixed before any merged-rating result was generated, is the main comparison; caps 7 and 9 test sensitivity to the choice of threshold. Confidence intervals for model differences use 10,000 paired bootstrap samples of whole song-title groups.

3.3. Comparison with expert courses

Dan-i Dojo is an official progression mode with increasingly advanced levels, each prescribing a three-chart course [16]. We use public community-maintained transcriptions of course tables for 2020–2025 [17]. The official mode description does not verify each transcribed row. Course placement was designed independently of our model. To compare equally rated charts, the main analysis retains only Oni charts rated 10, across 10th Dan, Kuroto, Meijin, Chojin, and Tatsujin, in increasing course level.

For this test, global one-sided Huber and sequence cumulative logit models, both of which learn the top rating as a lower bound, are trained on the original 1–10 scale. We freeze their test scores before matching chart and course records by exact normalized title. Ambiguous or unmatched titles are excluded. This yields 47 placements across 6 editions and 5 course levels. We compute Spearman correlation and pair accuracy within each year, comparing pairs from different course levels, then average the yearly values with equal weight. A 10,000-sample bootstrap resamples years, then placements within each year. A permutation test shuffles course levels within each year. As a control, we compare equally rated charts below 10. Position within a three-chart course is exploratory: only four main-analysis courses match all three positions, and position need not imply difficulty order.

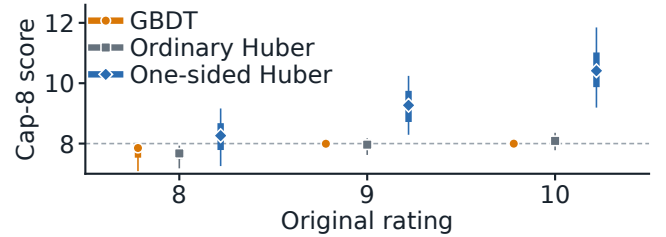


Fig. 2. Test scores of the 699 capped-group charts at cap 8 by original rating, for three global models trained on merged ratings. Markers: medians; thick bars: middle 50%; thin bars: 10th–90th percentiles. Dashed line: the cap.

4. RESULTS

4.1. Ranking after merging ratings

Table 1 shows that full-scale correlation and ranking within the capped group favor different models. Among seven global models, GBDT has the highest full-scale correlation ($\rho = 0.897$) but the lowest cap-8 correlation (0.435). Ordinal reduction is second on the full scale (0.890) and first at cap 8 (0.766), evaluated on the same 699 charts. The reversal also appears in Kendall correlation and pair accuracy. On the 159,318 unequal-rating pairs at cap 8, GBDT gives $\tau_b = 0.338$ and $A_8 = 0.709$; ordinal reduction gives 0.629 and 0.889, respectively.

4.2. Mechanism of the reversal

Figure 2 shows where capped-group scores fall relative to the cap. GBDT places 56% of capped-group test scores within 0.05 of the cap, and within this band their Spearman correlation with original ratings is 0.028 (95% CI [-0.075, 0.129]). Ordinary Huber places fewer scores there (16%), but they are

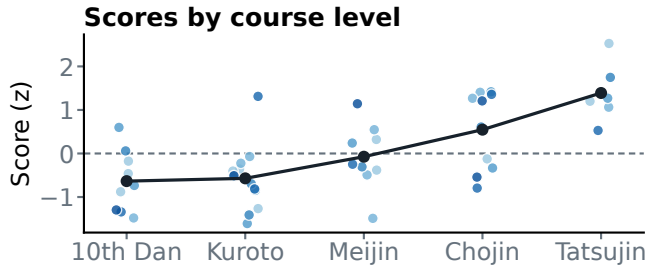


Fig. 3. Scores rise with course level among 10-rated charts (47 placements); models are trained on ratings 1–10. Global one-sided Huber scores, standardized within each year (darker blue: later year); black points are course-level means.

likewise unordered ($\rho = 0.008$ [-0.172, 0.184]). One-sided Huber places only 3% near the cap, and its scores rise above the cap with original rating. Concentration at the cap alone does not remove order: ordinal reduction places 60% of its scores within 0.05 of the cap, yet keeps them ordered ($\rho = 0.683$ [0.624, 0.735]). Exact targets thus pull capped charts toward the cap, where their scores carry little order; lower-bound and ordinal models keep that order.

4.3. Learning merged ratings as lower bounds

For global Huber regression, treating the cap as a lower bound improves Spearman correlation by +0.092 (95% CI [+0.058, +0.127]). Gains at caps 7 and 9 are +0.229 [+0.182, +0.277] and +0.044 [+0.014, +0.074]. Global ordinal models, which also impose no upper target, match or exceed the one-sided model. At every cap, the three lowest global correlations come from models that fit the cap as an exact target.

With the shared Transformer, ordinary Huber, one-sided Huber, and cumulative logit reach cap-8 correlations of 0.454, 0.671, and 0.718. The one-sided gain over ordinary Huber is +0.217 (95% CI [+0.180, +0.257]). With only the loss changed, the gain is attributable to the treatment of merged ratings.

Compared with global cumulative logit ($\rho = 0.761$), the sequence cumulative logit model differs in correlation by -0.043 (95% CI [-0.080, -0.006]). This comparison changes both input representation and architecture, so it compares complete systems without isolating the effect of temporal information. The sequence model does not outperform global cumulative logit. A larger corpus could change this ranking.

4.4. Agreement with expert-course levels

Figure 3 compares model scores with course levels among charts all rated 10. Global one-sided Huber scores have mean yearly Spearman correlation $\rho = 0.615$ (95% CI [0.248, 0.827]) and mean yearly pair accuracy 0.767 [0.605, 0.895] from 152 eligible pairs, a mean of yearly accuracies, not a pooled fraction. Both permutation tests give $p < .001$.

The sequence cumulative logit model agrees in direction, with mean yearly $\rho = 0.504$ [0.101, 0.756] and pair accuracy 0.709 [0.532, 0.859]. In the control analysis of 201 matched Oni placements, global scores correctly order a fraction 0.626 of the 425 within-year pairs sharing a rating below 10. This weaker control is descriptive because course composition and the distribution of ratings vary across course levels.

5. DISCUSSION AND CONCLUSION

Correlation with published ratings and correct ranking within their highest category are separate properties; GBDT’s reversal shows that the second needs its own evaluation. Comparisons across model families describe this dataset, not a universal advantage for any estimator.

Agreement with expert courses supports using scores to suggest smaller progression steps or flag disagreements for content review within an official rating. The scores express relative order and are not a calibrated extension of the star scale: a score of 13 does not mean three stars above 10. Official ratings remain the primary labels.

Original ratings provide the evaluation order in the artificial-cap test; they are not psychophysical ground truth. Course levels likewise reflect design choices, including pattern coverage, novelty, and the other charts in a course. The course comparison contains 47 placements across six editions; exact matching excludes three otherwise eligible placements. The course comparison covers two models, so the model reversal is tested only under artificial caps. The study covers one game and one expert chart category. It evaluates recovery of rating order and agreement with course design; it does not measure player performance or establish generalization to other domains.

Merging known levels creates a direct test of ranking at a ceiling, and ceiling labels should be learned as lower bounds. Applying the procedure elsewhere requires an ordinal scale whose upper levels can be withheld, or an external source of order; a real ceiling with neither cannot be validated from its shared rating alone. Rankings within a shared top rating can, and should, be evaluated wherever upper levels can be withheld or an external order exists.

Acknowledgments. Anthropic Claude and OpenAI GPT translated, edited, and revised text in all sections, supported discussion of the content, and, as instructed by the author, wrote the experiment code and the code for visualizations. The author verified all outputs and is responsible for the content.

Compliance with Ethical Standards

This study analyzes no human participants, player records, interactions, or interventions; its external criterion is publicly documented course placement. This study received no external funding, sponsorship, or donated computing resources. The author has no conflicts of interest to disclose.

6. REFERENCES

- [1] Bandai Namco Entertainment Inc., “Taiko no Tatsujin: Song list,” <https://taiko.namco-ch.net/taiko/en/songlist/>, Accessed July 2026.
- [2] Caroline B. Terwee, Sandra D. M. Bot, Michael R. de Boer, Daniëlle A. W. M. van der Windt, Dirk L. Knol, Joost Dekker, Lex M. Bouter, and Henrica C. W. de Vet, “Quality criteria were proposed for measurement properties of health status questionnaires,” *Journal of Clinical Epidemiology*, vol. 60, no. 1, pp. 34–42, 2007.
- [3] Bingheng Li and Fushuo Huo, “REQA: Coarse-to-fine assessment of image quality to alleviate the range effect,” *Journal of Visual Communication and Image Representation*, vol. 98, 2024, Art. no. 104043.
- [4] Yudai Tsujino, Ryosuke Yamanishi, and Yoichi Yamashita, “Characteristics study of dance-charts on rhythm-based video games,” in *2019 IEEE Conference on Games*, 2019, pp. 1–4.
- [5] Arturo P. Caronongan, III and Nelson A. Marcos, “Predicting chart difficulty in rhythm games through classification using chart pattern derived attributes,” in *Computational Science and Technology*, vol. 724 of *Lecture Notes in Electrical Engineering*, pp. 193–205. Springer, Singapore, 2021.
- [6] Billy Joe Franks, Benjamin Dinkelmann, Marius Kloft, and Sophie Fellenz, “Ordinal regression for difficulty prediction of StepMania levels,” in *Machine Learning and Knowledge Discovery in Databases: Applied Data Science and Demo Track*, 2023, pp. 497–512.
- [7] Pedro Ramoneda, Nazif Can Tamer, Vsevolod Eremenko, Xavier Serra, and Marius Miron, “Score difficulty analysis for piano performance education based on fingering,” in *2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022, pp. 201–205.
- [8] Peter McCullagh, “Regression models for ordinal data,” *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 42, no. 2, pp. 109–127, 1980.
- [9] Ling Li and Hsuan-Tien Lin, “Ordinal regression by extended binary classification,” in *Advances in Neural Information Processing Systems*, 2006, vol. 19, pp. 865–872.
- [10] Pedro A. Gutiérrez, María Pérez-Ortiz, Javier Sánchez-Monedero, Francisco Fernández-Navarro, and César Hervás-Martínez, “Ordinal regression methods: Survey and experimental study,” *IEEE Transactions on Knowledge and Data Engineering*, vol. 28, no. 1, pp. 127–146, 2016.
- [11] Kostiantyn Antoniuk, Vojtěch Franc, and Václav Hlaváč, “Interval insensitive loss for ordinal classification,” in *Proceedings of the Sixth Asian Conference on Machine Learning*. 2015, vol. 39 of *Proceedings of Machine Learning Research*, pp. 189–204, PMLR.
- [12] Naresh Manwani, “PRIL: Perceptron ranking using interval labels,” in *Proceedings of the ACM India Joint International Conference on Data Science and Management of Data*, 2019, pp. 78–85.
- [13] Vlad-Rareș Dănilă and Cătălin Buiu, “A deep learning approach to censored regression,” *Pattern Analysis and Applications*, vol. 27, 2024, Art. no. 24.
- [14] James Tobin, “Estimation of relationships for limited dependent variables,” *Econometrica*, vol. 26, no. 1, pp. 24–36, 1958.
- [15] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems*, 2017, vol. 30.
- [16] Bandai Namco Entertainment Inc., “Dan-i Dojo,” https://taiko.namco-ch.net/taiko/en/special/dani_dojo/index.php, Accessed August 2026.
- [17] Taiko Wiki Contributors, “Nijiro Dan-i Dojo course tables, 2020–2025,” <https://taiko.wiki/dani/25?lang=en>, Community-maintained transcription; select editions 2020–2025 using the year menu; accessed August 2026.